Keeping Bias Out of Lending AI

What It Takes to Audit Consumer Credit and Mortgage Origination, and Where the Standard Toolchain Leaves You

Siddhant Jain
Simon Fraser University
Burnaby, British Columbia, Canada

Ayaan Lalani
Simon Fraser University
Burnaby, British Columbia, Canada

Nicholas Vincent
Simon Fraser University
Burnaby, British Columbia, Canada

ABSTRACT

Lenders are deploying models into two decisions that shape household wealth: consumer installment credit and residential mortgage origination. Both fall under the Equal Credit Opportunity Act and Regulation B; mortgage adds the Fair Housing Act and, through HMDA, public reporting. The question this paper asks is operational rather than definitional: what does a team actually have to do to keep bias out of a model in these sectors, and how far does the standard toolchain carry them?

We built a complete audit for both sectors on real data — 1,000 consumer credit applications and 10,978 Georgia mortgage applications — with a deterministic pipeline computing every metric and a language model restricted to interpretation. We then measured what it took to make that audit correct.

Two results concern the standard toolchain, and we verify them directly against AIF360 and Fairlearn rather than inferring them. AIF360 ships `favorable_label=1.0` as a default, so on a target named `loan_default` it treats defaulting as the favorable outcome: the default returns a disparate impact of 1.2072 where correct orientation returns 0.9862. Fairlearn exposes no favorable-label parameter at all. And neither library has any concept of minimum subgroup size — no floor, no warning, no guidance — so every team invents its own.

The remaining results concern choices we had to make because nothing supplied them, and we report them as evidence about building such an audit rather than as properties of the field. Auditing a 2,196-row sample, our pipeline ranked American Indian men as the *best*-treated group in the mortgage data (DI = 1.1504) on the evidence of four applicants who all happened to be approved; over all 10,978 they are the worst group in the data (**0.6734**). Small samples invert disparities rather than merely hiding them, and a size floor cannot distinguish the two cases. We audited age at 40+, a threshold from employment law, and the model looked clean (0.9791); on the cut credit law specifies, 62+, it shows the largest age disparity in the data (**0.8897**).

We close with a six-step protocol for these two sectors, and with the observation that every failure above produced a *clean* result — the direction least likely to prompt a second look.

CCS CONCEPTS

• **Social and professional topics** → **Governmental regulations**; • **Computing methodologies** → *Machine learning*; • **Human-centered computing** → *Human computer interaction (HCI).*

KEYWORDS

algorithmic auditing, fairness measurement, disparate impact, intersectionality, lending, ECOA, large language models

1 INTRODUCTION

Credit decisions are among the most consequential automated decisions people encounter, and two of them dominate household outcomes: consumer installment credit, which governs access to short-term liquidity, and residential mortgage origination, which governs access to the principal asset most households will ever own. Both are directly regulated. The Equal Credit Opportunity Act and Regulation B govern both; the Fair Housing Act adds to mortgage; and the Home Mortgage Disclosure Act makes mortgage lending publicly auditable in a way almost no other consumer decision is.

Models are being deployed into both. The question we take up is not whether bias matters in these settings — that is settled, legally and empirically [2, 12, 31] — but what a practitioner concretely has to *do* to keep it out, and how much of that work the standard fairness toolchain performs for them. Lee et al. [22] make the organising commitment of human-centered fair AI that fairness is defined relative to a human decision context rather than in metric space. In lending, that decision context is written down: it is a statute. So the question becomes tractable in a way it is not in most domains — we can ask whether the tooling's defaults agree with the law.

This is not speculative. Deng et al. [10] find practitioners using fairness toolkits in ways their designers did not anticipate; Lee and Singh [23] map systematic gaps in what open-source toolkits support; Holstein et al. [17] report that practitioner needs diverge sharply from what the literature supplies. Mei et al. [28] show, in a speech setting, that audit *pipelines* introduce pitfalls that change conclusions, and Zaccour et al. [39] show that quantitative audits are more often blocked by data access than by method. The common thread is that the instrument is part of the finding.

Scope. We work in two regulated micro-sectors and one adjacent setting. German Credit [16] stands in for consumer installment credit, HMDA Georgia [8] is residential mortgage origination, and a Lending Club peer-to-peer default task provides a contrasting case where the protected attribute is unavailable. The first two carry the regulatory regime this paper is about; the third is retained because it exhibits a metric failure independent of any protected class.

What we did. We built a complete fairness audit for these settings and then examined the audit rather than only its output. The deterministic layer computes every metric with AIF360 [3] and assigns every severity label by fixed rule; a language model supplies interpretation and is scored against that fixed ground truth, so no model grades another model's opinion. Audits run over the full cleaned population — 1,000 and 10,978 rows — with every row scored out-of-fold by a model that never saw it, rather than over a held-out split.

What we found. Six components of an ordinary audit stack changed its conclusions (Table 1). Each is defensible in isolation. Each resolves a normative question without recording that it did so.

Contributions. (i) An empirical demonstration, on real lending data, that four standard components of an audit stack — a reporting floor, a protected-class threshold, a label convention, and two output detectors — each change what the audit concludes. (ii) A demonstration that small-sample audits invert disparities rather than only hiding them, isolated by holding the model fixed and varying only the audited population. (iii) A cross-model result showing that the prompt-sensitivity finding widely reported for LLM evaluators is specific to a model generation. (iv) A released, cost-capped, offline-reproducible audit pipeline in which every claim above is regenerable.

2 RELATED WORK

Auditing, and auditing the auditors. Metaxa et al. [29] survey algorithm auditing as method; Raji and Buolamwini [32] show external audits change vendor behaviour, and Raji et al. [33] give an internal end-to-end framework. Kroll [21] argues traceability is what makes accountability operational — the stance our frozen-artifact design takes. Documentation interventions [15, 30] and process interventions [27] share our premise that fairness work needs infrastructure, and share the exposure we report: a checklist is a mechanical proxy for judgement, and §5.4 is a case study in what that costs. Mei et al. [28] is the nearest prior work; Zamfirescu-Pereira et al. [40] diagnose human error inside an image-analysis pipeline. We add a lending case where the pitfalls are a size threshold and a statutory misattribution, and where the affected quantity is which group the audit names as most harmed.

Measurement. That fairness criteria conflict is settled [7, 20], and that choosing among them is not a technical act is argued by Selbst et al. [35], whose abstraction traps describe our setting. Corbett-Davies and Goel [9] catalogue how fairness measures mislead detached from decision context. Barocas and Selbst [1] and Wachter et al. [37] establish that legal discrimination does not reduce to a metric threshold, which is why we treat the four-fifths screen as a screening signal and make no legal claim. On intersectionality, Buolamwini and Gebru [5] give the canonical empirical demonstration that marginal accuracy hides joint-group failure and Foulds et al. [14] the formal definition; §5.1 contributes the observation that a routine size floor removes exactly those joint groups from the report.

Whose criteria. A growing line asks who should choose fairness criteria rather than how to compute them: Luo et al. [25] develop a protocol for negotiating metrics with stakeholders, Luo et al. [26] document how much stakeholders disagree on identical evidence, Luo et al. [24] extend this to affected individuals, and Deshpande and Sharp [11] ask who the stakeholders are at all. Stumpf et al. [36] argue fairness assessment needs user-centred evaluation; Kim et al. [19] find the construct of equity fragmenting under practical demand. Our contribution is adjacent: this literature asks who sets the criteria, and we show that the apparatus resolves several such questions before anyone is asked.

Language models as evaluators. Using an LLM to interpret metrics invites the LLM-as-judge critique [41]. Our design avoids the worst of it: ground truth is a deterministic rule over computed metrics, never another model's opinion. Wester et al. [38] study when and how LLMs deny user requests, which is precisely the phenomenon our refusal detector was built to catch and, as §5.4 shows, failed to measure.

3 WHAT THE STANDARD TOOLCHAIN DECIDES FOR YOU

Before reporting our own results we establish what the two most widely used fairness libraries do by default, since the rest of this paper depends on the distinction between their behaviour and our choices. We inspected AIF360 0.6 [3] and Fairlearn 0.13 [4] directly.

Which outcome counts as favorable. AIF360's `BinaryLabelDataset` takes `favorable_label` with a default of 1.0. Credit risk targets are conventionally coded with the adverse event as 1 — default, delinquency, charge-off, fraud — so on a column named `loan_default` the default asserts that defaulting is the good outcome. A minimal reproduction on synthetic data where women default slightly more often returns DI = 1.2072 under the library default and 0.9862 under correct orientation: the default reports the disadvantaged group as *favoured.* Fairlearn's `demographic_parity_ratio` exposes no favorable-label parameter at all and computes on the positive class implicitly. Nothing in either API requires the practitioner to state which outcome is good, and nothing warns when the choice is left implicit.

How much data a subgroup needs. Neither library has any concept of minimum subgroup size. There is no floor, no warning, and no guidance; Fairlearn offers `count` as an opt-in metric a user may choose to request. This is the gap §5.1 concerns, and it matters because the decision is therefore made privately by every team that performs an intersectional audit.

Who counts as protected. Neither library binds a protected attribute to a statutory definition. The practitioner passes a column; nothing checks it against ECOA, and nothing records which reference group produced a ratio.

These libraries compute what they are asked to compute, correctly, and we are not reporting bugs. The observation is narrower: four questions that determine whether a lending audit is legally meaningful — which outcome is favorable, how much data a subgroup needs, who counts as protected, and against whom — are resolved either by a silent default or not at all.

Table 1: Results and the evidence for each. Every artifact path is in the released repository; every figure is regenerated by the pipeline rather than transcribed.

Finding	Evidence	Proof
Small samples inverted a disparity	On a held-out split, `American Indian × Male` reads 1.1504 — above parity, the best cell — on four applicants. At full population the same group is **0.6734**, the worst cell in the data. Small samples invert disparities, not merely hide them.	`scripts/compare_populations.py`; `population_comparison.json`
The age audit used the wrong statute	40+ is the ADEA (employment). ECOA protects 62+ (Reg. B §1002.2(o)). Pooling young against senior applicants cancels them: banded `age_group` reads 0.9791 (LOW); the statutory cut reads **0.8897**, the largest age disparity in the data.	`hmda/fairness/fairness_summary.json`; `clean_hmda.py`
A label convention inverted a headline metric	Selection rates computed on the unfavorable label of a target named `loan_default` gave DI = 0.0 (CRITICAL) where the correct value is 0.9931.	`tests/test_lending_club_fairness.py`
A refusal detector measured brevity, then set policy	6 flagged responses, 0 genuine refusals: a 22-character correct answer fell under a 40-character threshold, under a prompt demanding terseness. Applied to a second provider it zeroed 14 of 24 cycles, moving its mean from 70.0 to 19.79; that provider is permanently blocked in our config on that evidence.	`section_f/gemini_hallucination_summary.json`; `research_guardrails.json`
A fabrication detector penalised arithmetic	`gpt-5-mini` tripped 5× more flags than `gpt-4o` on identical input, but 14 of 21 (67%) flagged values are derivable from the provided context (88.88 is 0.8888 as a percentage).	`llm_benchmark/gpt-5-mini/`; `RESULTS.md` §4.1
Prompt sensitivity is model-generation-specific	Cycle-mean spread across four prompt strategies: 14.7 and 9.0 points for `gpt-4o`; 0.0 and 1.9 for `gpt-5-mini`, same packs.	`llm_benchmark_v2/gpt-4o/`; `RESULTS.md` §4

4 WHAT WE BUILT

4.1 Three lending decisions

The three settings differ along the two axes that most change what an audit can conclude: whether the protected attribute is a protected class or an economic characteristic, and whether the classifier discriminates enough for disparity to be measurable. We use German Credit [16], HMDA Georgia [8], and a Lending Club default task. Lending discrimination is documented empirically across this span — appearance and gender effects in peer-to-peer markets [12, 31], FinTech-era consumer lending [2], local credit-market conditions [6], and historical records used as training data [34].

4.2 Deterministic ground truth

For each protected attribute we compute disparate impact, demographic parity difference, equal opportunity difference, average odds difference and the Theil index with AIF360 [3], cross-checked against Fairlearn [4]. A fixed rule maps disparate impact to a severity band, with the CRITICAL boundary at 0.72 and the four-fifths screen at 0.80. The model never recomputes a number; asserting an unprovided value is recorded as fabrication, not as a result.

4.3 Division of labour: deterministic numbers, model interpretation

A fairness metric is not a finding. Turning DI = 0.8897 into something a credit committee can act on requires naming who is affected, what mechanism is plausible, what the applicable rule is, and what remediation is available at what cost. That interpretive step is real work, and it is what language models are good at — which is why they are being wired into compliance tooling.

The evaluation problem is that interpretation has no ground truth unless something outside the model supplies it. Our architecture supplies it by construction, and the constraint is strict:

(1) **Python owns every number.** Metrics come from AIF360 [3]; severity comes from a fixed rule over those metrics. The model receives them as given context.
(2) **The model owns only the narrative.** It never recomputes a metric. A numeric value in its output that is absent from the provided context is recorded as fabrication, not as a result.
(3) **Scoring is against the deterministic label**, never against another model's opinion — which sidesteps the central weakness of model-graded evaluation [41].

Calculator, not oracle. The practical payoff is that when a model and the pipeline disagree, the disagreement is attributable: the numbers did not move.

4.4 Two benchmark tracks, kept separate

The two questions one might ask of an LLM auditor are different and we do not let them contaminate each other.

Track Q — is it a reliable interpreter? For each (use case, attribute) pair we issue four prompt strategies — `zero_shot`, `chain_of_thought`, `self_critique`, `constrained` — and score the returned severity against the deterministic label. All prompts are frozen and released, and a replay path reproduces the entire scoring pipeline with no API key.

Table 2: Marginal disparate impact, full population. Read alone, this table reports an unremarkable audit. †The ECOA cut; `age_group` is the banded view of the same applicants. ‡Synthetic attribute (§4).

Setting	Attribute	DI
German Credit	`Sex`	0.8888
	`AgeGroup`	0.9258
	`foreign_worker`	0.8387
HMDA (Georgia)	`race`	0.9371
	`sex`	0.9572
	`age_group` (banded)	0.9791
	`age_62_plus`†	**0.8897**
Lending Club	`gender`‡	0.9931
	`income_level`	1.0072
	`loan_amount_level`	1.0063

Track H — does it raise the audit-quality ceiling? Narrative audits, one per use case, judged only against a five-criterion checklist, a guardrail baseline and a quality rubric — never against the deterministic severity numbers, which would merely re-run Track Q.

Spend is capped cumulatively with a pre-call abort and a persisted ledger: 98 calls, \$0.8933, against a \$15 cap.

4.5 Auditing the population, not a sample

Audits here run over every cleaned row, with predictions generated by stratified five-fold cross-validation so each applicant is scored by a model that never saw them. This matters more than it sounds: at held-out-split size the HMDA `race×sex` cells for American Indian, Multiracial and Pacific Islander applicants hold three to six people each. Those groups do not become measurable through a better estimator; they become measurable through more rows.

4.6 Disclosure

None of the three models would pass a real fair lending review, for reasons orthogonal to our claims, and we state this rather than let it be discovered: the German Credit model takes applicant sex and age as input features; the Lending Club `gender` attribute is synthetic, assigned by an independent coin flip, so that setting contains no protected class; and HMDA's public data carry no credit score, the primary underwriting variable. We audit these systems as instruments for studying audit infrastructure, not as candidate lending models.

5 RESULTS

We report marginal disparate impact for every protected attribute in Table 2 as the baseline against which the rest of this section should be read. On these numbers alone, the audit is unremarkable: every attribute in every setting clears or nearly clears the four-fifths screen, and nothing demands attention. Each subsection below concerns a component of the audit that determined one of these numbers, or determined which numbers appeared at all.

Table 3: HMDA `race` × `sex` under an $n \geq 15$ floor. Identical model; only the audited population differs.

	Held-out split (n=2,196)	Full population (n=10,978)
Cells suppressed	**5 of 11**	1 of 12
Worst cell *reported*	0.8169 (MOD.)	**0.6734** (CRIT.)
`Am. Indian×Male`	**1.1504** (best cell)	**0.6734** (worst cell)
`Am. Indian×Female`	0.1917 (n=6, noise)	0.8815 (n=25)

5.1 Small samples invert disparities, and the floor cannot help

Subgroup reporting floors are standard and defensible: a ratio computed on six people is not evidence. Ours suppresses cells below $n = 15$. To isolate what that costs, we hold the model fixed and vary only the audited population (Table 3).

The failure is sharper than suppression. On the split, `American Indian × Male` appears at DI = 1.1504 — *above* parity, the best-treated cell in the table. Four applicants, four approvals. At full population that same group is at 0.6734 with n=31: the worst cell in the dataset, in the CRITICAL band. The audit did not miss the disparity; it **inverted** it, and attached a reassuring number to the inversion.

The floor was right to suppress what it suppressed. `American Indian × Female` reads 0.1917 on the split — one approval in six — which looks like the most severe disparity in the study and is in fact noise: at full population the same cell is 0.8815 (n=25). A reader who overrode the floor to surface that number would have reported a catastrophe that does not exist.

That is the uncomfortable shape of this result. The floor is not the error and removing it is not the fix. At this sample size the estimates for small groups are unreliable *in both directions* — one group looked catastrophic and another looked flawless, and both were wrong. Every affected cell belonged to a racial minority, which is mechanism rather than coincidence: a floor on group size is a floor on group population, so the groups whose treatment is least certain are exactly the groups least likely to be counted.

At full population the same floor suppresses one cell of twelve and the worst disparity, `American Indian × Male` at 0.6734 (n=31), becomes visible. Table 4 gives the complete cell table, which we print in full because the argument of this paper is precisely that partial tables mislead. The three worst cells hold 31, 18 and 29 applicants and would be invisible under the split-sized audit; the two cells an audit typically discusses, `Black × Female` and `Black × Male`, sit sixth and seventh.

The floor is not a bug and we do not propose removing it. The point is narrower and harder to design around: a threshold chosen for statistical prudence distributes its entire cost onto the smallest groups, which in lending are generally the groups least able to absorb it, and it does so silently — the audit output lists what passed, not what was withheld. Our own codebase carried two different floors, $n \geq 15$ and $n \geq 30$, in adjacent modules, so the same cell was reportable or not depending on which one read it.

Table 4: HMDA race × sex, full population. The floor suppresses the starred row. On a held-out split it suppressed the top five.

race	sex	*n*	sel. rate	DI
American Indian	Male	31	0.5806	**0.6734**
Multiracial	Female	18	0.6111	0.7088
Multiracial	Male	29	0.6552	0.7599
Pacific Islander	Female	12*	0.6667	0.7732
Pacific Islander	Male	15	0.7333	0.8505
Black	Female	1,994	0.7442	0.8632
Black	Male	1,707	0.7452	0.8643
American Indian	Female	25	0.7600	0.8815
White	Female	2,318	0.7865	0.9121
White	Male	4,019	0.8281	0.9604
Asian	Male	556	0.8525	0.9888
Asian	Female	254	0.8622	1.0000

5.2 The age audit used the wrong statute

Fairness work on credit routinely audits age at 40+, and ours did. That threshold is from the Age Discrimination in Employment Act, which governs employment. Credit is governed by the Equal Credit Opportunity Act, whose Regulation B §1002.2(o) defines the protected class as **62 or older**. A comment in our own cleaning code asserted that 40+ "aligns with US ECOA". It does not.

The substitution is not cosmetic. Our banded age attribute pools young and senior applicants against the middle band; the two move in opposite directions, so pooling cancels them. The banded attribute returns DI = 0.9791, severity LOW — a clean bill of health. Audited on the statutory cut, the same model over the same rows returns **0.8897**, the largest age disparity in the dataset and the lowest of its four protected attributes. The flag required, `applicant_age_above_62`, was present and unused in the raw HMDA file; it is published precisely because §1002.2(o) turns on it, and it is necessary because the public age bands straddle 62, so no banded attribute can isolate the protected class.

One asymmetry this exposes and cannot resolve: §1002.6(b)(2) expressly permits *favouring* an elderly applicant. Disparate impact is symmetric, so a ratio above 1 here is lawful and is not a reverse-disparity finding, yet the metric reports it identically. The instrument cannot express the rule it is being used to check.

5.3 A label convention inverted a headline metric

In the Lending Club setting, `gender` initially returned DI = 0.0 and a CRITICAL label — the most severe output the pipeline can produce. It was an artifact. The target is `loan_default`, so label 1 is the *unfavorable* outcome; selection rates computed on predicted default give 0.0000 and 0.0069, hence DI = 0.0. Oriented on the favorable outcome the rates are 1.0000 and 0.9931, so DI = 0.9931: near parity.

Nothing was misused. AIF360 computed what it was asked. The data were clean. Disparate impact was the right screen. The error lives entirely in the mapping from a column name to a normative category, and that mapping is a convention rather than a computation: a target named `approved` makes label 1 favorable, a target named `loan_default` inverts it, and nothing in the metric signature records which case holds. Default, delinquency, charge-off and fraud are all conventionally coded with the adverse event as 1, so fairness tooling applied to any risk model inherits this unless an analyst supplies the mapping. It is now a required argument and a regression test in our pipeline, which converts a silent inversion into a build failure.

5.4 Automated output checks penalise correct behaviour

Where a language model sits in the audit loop, its output must be checked automatically or the pipeline does not scale. We implemented the two checks such a system obviously needs — a refusal detector and a fabricated-metric detector — and measured what they actually score.

The refusal detector measured brevity. It flagged six of nine responses under a terse-output prompt strategy. None was a refusal: no flagged record contained a refusal phrase, a missing field, or an invalid severity. The detector treats any narrative field under 40 characters as effectively empty, so `how_to_fix: "DisparateImpactRemover"` — correct, on-spec, and naming the applicable post-processor [13] in 22 characters — registered as a refusal. Compliance with the instruction was the trigger. Reporting the raw output would have yielded a 66.7% refusal rate and a tidy story about constrained prompting inducing evasion, entirely an artifact of our instrument.

And then it set policy. Applied to a second provider, the same detector flagged 19 of 24 cycles and mechanically zeroed 14 of them. Every zero-score cycle is a flagged cycle; excluding them moves that provider's mean score from 19.79 to **70.0**. At least five flagged cycles are not refusals — one scored 85/100. On the strength of the 19.79 figure, that provider was recorded as permanently blocked in our guardrail configuration, where the block persists. An invalid measurement did not merely mislead a table; it produced a durable governance decision about a vendor.

The fabrication detector penalised arithmetic. Our second detector flags any numeric value absent from the provided context. Replaying identical packs against two model generations, `gpt-5-mini` trips 17 flags across 28 calls where `gpt-4o` trips 4. Read naively, the newer model fabricates five times as often. Checking each flagged value against the context, **14 of 21 (**67%**) are derivable from it**: 88.88 is the disparate impact 0.8888 restated as a percentage, 83.87 is 0.8387, and most of the remainder are gaps between two provided values expressed in percentage points. The detector matches raw float literals, so converting a ratio to a percentage or subtracting two given numbers registers as fabrication. The model that reasons more explicitly is ranked as the less trustworthy one, precisely because it shows its work.

A guardrail punished "collect more data". A third check requires every proposed mitigation to name a canonical algorithm — reweighing [18], disparate impact removal [13], and similar. It failed exactly the two attributes whose problem *is* data scarcity, where the audit proposed acquiring records to a stated floor, hierarchical shrinkage and manual review. No post-processor repairs a six-record subgroup. The gate encodes an assumption that every fairness problem has an algorithmic mitigation, and where that is false it penalises the only defensible answer.

All three checks are left as implemented, and their measured behaviour reported alongside their intended behaviour. The generalisable result is that each mechanical check encodes a substantive judgement — what counts as an answer, what counts as evidence, what counts as a remedy — and each misfires on the inputs where that judgement is hardest: the terse-but-correct answer, the model that shows its arithmetic, the subgroup too small for any algorithm to repair. Automated evaluation inherits these judgements whether or not it records them.

5.5 Benchmark: what the model layer can and cannot be trusted with

Aggregate agreement is not a capability measure. Severity agreement between `gpt-4o` and the deterministic labels is 27/36 = 75.0%. It decomposes to 33.3%, 91.7% and 100.0% across the three settings, and the ordering tracks how discriminating each audited classifier is rather than how capable the model is: the 100% case is one where the classifier barely separates anyone, so every attribute is a trivial near-parity LOW and agreeing costs nothing. A single headline agreement figure for an LLM auditor is therefore not a well-defined quantity. Errors are asymmetric in the safe direction — seven over-escalations against two under-escalations — which matters because under-escalation is the consequential failure for an audit tool.

Grounding beats prompt engineering. Embedding retrieved literature in the prompt, holding model and attributes fixed, cut fabricated citations from 16 to 2 and raised mean quality from 69.08 to 76.67. This is the largest single effect in the benchmark, and it is a design result rather than a model result: it is available to anyone building such a system.

Narrative audits clear a mechanical quality bar. Track H audits verified 27/27 citations against the retrieved evidence pool, with weighted rubric scores of 4.8, 4.7 and 5.0 out of 5 (cross-model and reported as secondary, since a model grading a model is weaker than a mechanical gate). Two behaviours the deterministic template cannot produce: on a six-record subgroup the audit independently reproduced the small-n caveat *and* quantified it — one label change would swing the ratio to 0.78, therefore power is inadequate — and on a near-degenerate classifier it declined the flattering reading, converting a null into a conditional prediction with a mechanism.

5.6 Prompt sensitivity is model-generation-specific

A widely reported result about LLM evaluators is that prompt phrasing moves the verdict. It reproduces here, and it does not generalise across model generations. On identical packs, the spread in cycle-mean score across four prompt strategies is 14.7 and 9.0 points for `gpt-4o` and **0.0** and **1.9** for `gpt-5-mini`. Claims of the form "LLM auditors are unstable under paraphrase" should therefore carry a model and a date.

Two methodological notes. `gpt-5-mini` rejects the `temperature` parameter outright, so a temperature sweep is impossible on it and any repeated-sampling study must use another model — an API constraint that silently determines which experiments a benchmark can run. And our four-cycle design varies prompt strategy rather than resampling one strategy, so cycle-to-cycle differences confound phrasing with sampling noise; the cross-model contrast above is robust to this, since both models face the same confound.

6 WHY THIS MATTERS

6.1 The gap is between what the tooling checks and what the law requires

The two findings in §3 are about widely used libraries and hold for anyone who uses them. The rest of our results are about choices we made because those libraries make no choice — and that is precisely the point. A lending audit is legally meaningful only if it answers four questions correctly: which outcome is favorable, who counts as protected, how much data supports each subgroup estimate, and against whom the comparison runs. The toolchain answers the first by default and the remaining three not at all, so each is settled privately, by whoever writes the harness, without a record.

Both failure modes we document empirically — a threshold drawn from the wrong statute, and a sample too small for the affected subgroups — produce *clean* audit results. Neither trips any check the toolchain provides, because the toolchain provides none, and a clean result is the outcome least likely to be re-examined.

The asymmetry that matters is directional. A false alarm costs reviewer time. A false clearance closes an open question, and §5.1 and §5.2 are both false clearances — exactly the direction least likely to prompt a second look.

6.2 A single benchmark number can be an artifact of task difficulty

Our headline agreement figure between model and ground truth was 75%. It decomposes to 33.3%, 91.7% and 100%, and the 100% comes from the setting where the audited classifier barely discriminates at all, so agreeing costs nothing. The ordering tracks how hard each case is, not how capable the model is. This generalises past fairness: any benchmark averaging over heterogeneous instances can report a number dominated by the difficulty mix of its constituents rather than by the capability it claims to measure.

6.3 Every mechanical proxy for judgement is a policy

A length threshold that treats brevity as refusal is a policy about what counts as an answer. A keyword list that treats "acquire more data" as a non-mitigation is a policy about what counts as a remedy. A float-literal match that treats a percentage as a fabrication is a policy about what counts as evidence. None was labelled a policy; each was a heuristic written to make an evaluation automatable, and each punished the better answer in the cases where judgement was most needed. As evaluation automates further — LLM judges, automated red-teams, compliance scoring — this class of failure scales with it.

6.4 A workable division of labour

The architecture that survived our evidence is narrow and, we think, transferable: deterministic code owns the numbers and the thresholds; the model owns the interpretation; and the interpretation is scored against the numbers rather than against another model. Calculator, not oracle. The pattern extends to any setting where a model is asked to interpret computed quantities under consequence, and it is the reason we can state which of our own results are artifacts — the ground truth did not move when the model did.

7 WHAT TO DO ABOUT IT

These follow directly from the results and are cheap to adopt.

(1) **Make the favorable outcome an explicit, required argument.** Never infer it from label values. One parameter converts a class of silent inversions into a build failure (§5.3).
(2) **Report what the floor withheld.** An audit should print suppressed cells with their *n* and their unreported ratio, not omit them. The audit in §5.1 was not wrong about any number it computed; it was wrong about what it did not print.
(3) **Bind protected-class definitions to the governing statute** in code, with the citation next to the threshold. A comment asserting the wrong statute survived review precisely because it looked authoritative (§5.2).
(4) **Treat every mechanical gate as a hypothesis to falsify**, starting with inputs where the correct answer is unusual: the terse answer, the tiny subgroup, the model that shows its work.
(5) **Audit the population, not a convenience sample.** Most of our small-cell instability came from auditing a held-out split rather than every applicant in the review period.
(6) **Record which reference group a ratio uses.** Comparing against the most-favoured cell rather than a designated control group moved one verdict across the four-fifths line, and nothing in the metric signature recorded which convention was in force.

8 WHAT THIS DOES NOT SHOW

Scale. Three datasets, two primary models. These are mechanism-level findings: we show that a class of failure occurs and how it operates, not how often it occurs in the wild. The mechanisms transfer even where the statistics do not — a convention default that inverts a metric is a class of defect, not an incident.

No human subjects. We make claims about what an audit reader would conclude without testing them on audit readers. That is the natural next study and the main gap between this work and the stakeholder line it builds on [24, 25].

No causal or legal claims. All metrics are associational; root-cause mapping is rule-based inference over metric values. ECOA, Regulation B and the four-fifths rule appear as decision context. A disparate impact ratio below 0.80 is a screening signal, not a finding of unlawful discrimination [1, 37].

Model limitations. The German Credit classifier uses protected attributes as features; the Lending Club protected attribute is synthetic; HMDA lacks credit score. Each is disclosed in §4 and each bounds what the corresponding audit can mean.

Sampling confound. Temperature and seed are unpinned, so within-model cycle differences mix prompt strategy with sampling noise.

9 CONCLUSION

We asked what a team must do to keep bias out of a model in consumer credit and mortgage origination. The honest answer is that the standard toolchain computes the metrics correctly and leaves the decisions that make those metrics legally meaningful to the practitioner — silently, in the case of the favorable-label default, and entirely, in the case of subgroup size, statutory definition, and reference group.

Two of those decisions are load-bearing and measurably so. Sample size decides whether the worst-treated racial group in a mortgage portfolio appears as the worst-treated or the best-treated: at n=4 the ratio reads 1.1504, at n=31 it reads 0.6734. Statutory threshold decides whether the age class credit law protects appears disadvantaged at all: 0.9791 under an employment-law cut, 0.8897 under the credit-law one. Both errors produce clean results, which in lending is the direction nobody re-examines.

The protocol in §7 is what we would have wanted before starting: state the favorable outcome, audit the book rather than a sample, put the statute next to the threshold, publish what you suppressed, record the reference group, and test each automated gate where the right answer is unusual. None of it is technically difficult. All of it is currently optional, which is the problem.

ACKNOWLEDGMENTS

We thank the maintainers of AIF360 and Fairlearn. Retrieved literature was obtained via the Semantic Scholar API.

REFERENCES

[1] Solon Barocas and Andrew D. Selbst. 2016. Big Data's Disparate Impact. *SSRN Electronic Journal* (2016). https://doi.org/10.2139/ssrn.2477899
[2] Robert P. Bartlett, Adair Morse, Richard Stanton, and Nancy Wallace. 2019. Consumer-lending discrimination in the FinTech Era. *Journal of Financial Economics* (2019). https://doi.org/10.1016/j.jfineco.2021.05.047
[3] Rachel K. E. Bellamy, Kuntal Dey, Michael Hind, Samuel C. Hoffman, Stephanie Houde, Kalapriya Kannan, Pranay Lohia, Jacquelyn Martino, Sameep Mehta, Aleksandra Mojsilović, Seema Nagar, Karthikeyan Natesan Ramamurthy, John Richards, Diptikalyan Saha, Prasanna Sattigeri, Moninder Singh, Kush R. Varshney, and Yunfeng Zhang. 2019. AI Fairness 360: An Extensible Toolkit for Detecting and Mitigating Algorithmic Bias. *IBM Journal of Research and Development* 63, 4/5 (2019). https://doi.org/10.1147/JRD.2019.2942287
[4] Sarah Bird, Miro Dudík, Richard Edgar, Brandon Horn, Roman Lutz, Vanessa Milan, Mehrnoosh Sameki, Hanna Wallach, and Kathleen Walker. 2020. *Fairlearn: A toolkit for assessing and improving fairness in AI.* Technical Report MSR-TR-2020-32. Microsoft.
[5] Joy Buolamwini and Timnit Gebru. 2018. Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification. In *Proceedings of the 1st Conference on Fairness, Accountability and Transparency (Proceedings of Machine Learning Research).*
[6] Alexander W. Butler, Jess Cornaggia, and Umit G. Gurun. 2017. Do Local Capital Market Conditions Affect Consumers' Borrowing Decisions? *Management Science* (2017). https://doi.org/10.1287/mnsc.2016.2560
[7] Alexandra Chouldechova. 2017. Fair Prediction with Disparate Impact: A Study of Bias in Recidivism Prediction Instruments. *Big Data* 5, 2 (2017), 153–163. https://doi.org/10.1089/big.2016.0047

[8] Consumer Financial Protection Bureau. 2017. Home Mortgage Disclosure Act (HMDA) Loan/Application Records. https://www.consumerfinance.gov/data-research/hmda/.

[9] Sam Corbett-Davies and Sharad Goel. 2018. The Measure and Mismeasure of Fairness: A Critical Review of Fair Machine Learning. arXiv preprint.

[10] Wesley Hanwen Deng, Manish Nagireddy, Michelle Seng Ah Lee, Jatinder Singh, Zhiwei Steven Wu, Kenneth Holstein, and Haiyi Zhu. 2022. Exploring How Machine Learning Practitioners (Try To) Use Fairness Toolkits. In *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency (FAccT '22)*. 473–484. https://doi.org/10.1145/3531146.3533113

[11] Advait Deshpande and Helen Sharp. 2022. Responsible AI Systems: Who Are the Stakeholders?. In *Proceedings of the 2022 AAAI/ACM Conference on AI, Ethics, and Society (AIES '22)*. 227–236. https://doi.org/10.1145/3514094.3534187

[12] Jefferson Duarte, Stephan Siegel, and Lance A. Young. 2012. Trust and Credit: The Role of Appearance in Peer-to-peer Lending. *The Review of Financial Studies* (2012). https://doi.org/10.1093/rfs/hhs071

[13] Michael Feldman, Sorelle A. Friedler, John Moeller, Carlos Scheidegger, and Suresh Venkatasubramanian. 2015. Certifying and Removing Disparate Impact. In *Proceedings of the 21th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. https://doi.org/10.1145/2783258.2783311

[14] James R. Foulds, Rashidul Islam, Kamrun Naher Keya, and Shimei Pan. 2020. An Intersectional Definition of Fairness. In *2020 IEEE 36th International Conference on Data Engineering (ICDE)*. 1918–1921. https://doi.org/10.1109/ICDE48307.2020.00203

[15] Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daumé, III, and Kate Crawford. 2021. Datasheets for Datasets. *Commun. ACM* 64, 12 (2021), 86–92. https://doi.org/10.1145/3458723

[16] Hans Hofmann. 1994. Statlog (German Credit Data). UCI Machine Learning Repository. https://doi.org/10.24432/C5NC77

[17] Kenneth Holstein, Jennifer Wortman Vaughan, Hal Daumé, III, Miro Dudik, and Hanna Wallach. 2019. Improving Fairness in Machine Learning Systems: What Do Industry Practitioners Need?. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (CHI '19)*. 1–16. https://doi.org/10.1145/3290605.3300830

[18] Faisal Kamiran and Toon Calders. 2011. Data preprocessing techniques for classification without discrimination. *Knowledge and Information Systems* (2011). https://doi.org/10.1007/s10115-011-0463-8

[19] Seyun Kim, Yuanchen Bai, Haiyi Zhu, and Motahhare Eslami. 2025. A Systematic Literature Review on Equity and Technology in HCI and Fairness: Navigating the Complexities and Nuances of Equity Research. *Proceedings of the ACM on Human-Computer Interaction* (2025). https://doi.org/10.1145/3711079

[20] Jon Kleinberg, Sendhil Mullainathan, and Manish Raghavan. 2017. Inherent Trade-Offs in the Fair Determination of Risk Scores. In *8th Innovations in Theoretical Computer Science Conference (ITCS 2017)*. https://doi.org/10.4230/LIPIcs.ITCS.2017.43

[21] Joshua A. Kroll. 2021. Outlining Traceability: A Principle for Operationalizing Accountability in Computing Systems. In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency (FAccT '21)*. 758–771. https://doi.org/10.1145/3442188.3445937

[22] Min Kyung Lee, Nina Grgić-Hlača, Michael Carl Tschantz, Reuben Binns, Adrian Weller, Michelle Carney, and Kori Inkpen. 2020. Human-Centered Approaches to Fair and Responsible AI. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems (CHI EA '20)*. https://doi.org/10.1145/3334480.3375158

[23] Michelle Seng Ah Lee and Jat Singh. 2021. The Landscape and Gaps in Open Source Fairness Toolkits. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*. 1–13. https://doi.org/10.1145/3411764.3445261

[24] Lin Luo, Satwik Ghanta, Yuri Nakao, Mathieu Chollet, and Simone Stumpf. 2026. Empowering Affected Individuals to Shape AI Fairness Assessments: Processes, Criteria, and Tools. arXiv preprint arXiv:2602.06984. https://doi.org/10.48550/arXiv.2602.06984

[25] Lin Luo, Yuri Nakao, Mathieu Chollet, Hiroya Inakoshi, and Simone Stumpf. 2025. EARN Fairness: Explaining, Asking, Reviewing, and Negotiating Artificial Intelligence Fairness Metrics Among Stakeholders. *Proceedings of the ACM on Human-Computer Interaction* 9, 2 (2025), 1–37. https://doi.org/10.1145/3710908

[26] Lin Luo, Yuri Nakao, Mathieu Chollet, Hiroya Inakoshi, and Simone Stumpf. 2026. "I Think This Is Fair": Uncovering the Complexities of Stakeholder Decision-Making in AI Fairness Assessment. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26)*. 1–30. https://doi.org/10.1145/3772318.3790770

[27] Michael A. Madaio, Luke Stark, Jennifer Wortman Vaughan, and Hanna Wallach. 2020. Co-Designing Checklists to Understand Organizational Challenges and Opportunities around Fairness in AI. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)*. 1–14. https://doi.org/10.1145/3313831.3376445

[28] Katelyn X. Mei, A. S. G. Choi, Hilke Schellmann, Mona Sloane, and Allison Koenecke. 2025. Addressing Auditing Pitfalls in Automatic Speech Recognition Technologies: A Case Study of People with Aphasia. In *Proceedings of the 2025 ACM Conference on Fairness, Accountability, and Transparency*. https://doi.org/10.1145/3805689.3812320

[29] Danaë Metaxa, Joon Sung Park, Ronald E. Robertson, Karrie Karahalios, Christo Wilson, Jeff Hancock, and Christian Sandvig. 2021. Auditing Algorithms: Understanding Algorithmic Systems from the Outside In. *Foundations and Trends in Human–Computer Interaction* 14, 4 (2021), 272–344. https://doi.org/10.1561/1100000083

[30] Margaret Mitchell, Simone Wu, Andrew Zaldivar, Parker Barnes, Lucy Vasserman, Ben Hutchinson, Elena Spitzer, Inioluwa Deborah Raji, and Timnit Gebru. 2019. Model Cards for Model Reporting. In *Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT* '19)*. 220–229. https://doi.org/10.1145/3287560.3287596

[31] Devin G. Pope and Justin R. Sydnor. 2011. What's in a Picture? Evidence of Discrimination from Prosper.com. *The Journal of Human Resources* (2011). https://doi.org/10.3368/jhr.46.1.53

[32] Inioluwa Deborah Raji and Joy Buolamwini. 2019. Actionable Auditing: Investigating the Impact of Publicly Naming Biased Performance Results of Commercial AI Products. In *Proceedings of the 2019 AAAI/ACM Conference on AI, Ethics, and Society (AIES '19)*. 429–435. https://doi.org/10.1145/3306618.3314244

[33] Inioluwa Deborah Raji, Andrew Smart, Rebecca N. White, Margaret Mitchell, Timnit Gebru, Ben Hutchinson, Jamila Smith-Loud, Daniel Theron, and Parker Barnes. 2020. Closing the AI Accountability Gap: Defining an End-to-End Framework for Internal Algorithmic Auditing. In *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency (FAT* '20)*. 33–44. https://doi.org/10.1145/3351095.3372873

[34] Kanishk Sarwal and Sheikh Rabiul Islam. 2024. Algorithmic Lending Bias: Evaluating the Fairness of Historical Records. *IEEE International Congress on Big Data* (2024).

[35] Andrew D. Selbst, Danah Boyd, Sorelle A. Friedler, Suresh Venkatasubramanian, and Janet Vertesi. 2019. Fairness and Abstraction in Sociotechnical Systems. In *Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT* '19)*. 59–68. https://doi.org/10.1145/3287560.3287598

[36] Simone Stumpf, Evdoxia Taka, Yuri Nakao, Lin Luo, Ryosuke Sonoda, and Takuya Yokota. 2024. The Need for User-centred Assessment of AI Fairness and Correctness. In *Adjunct Proceedings of the 32nd ACM Conference on User Modeling, Adaptation and Personalization*. https://doi.org/10.1145/3631700.3664912

[37] Sandra Wachter, Brent Mittelstadt, and Chris Russell. 2021. Why Fairness Cannot Be Automated: Bridging the Gap Between EU Non-Discrimination Law and AI. *Computer Law & Security Review* 41 (2021), 105567. https://doi.org/10.1016/j.clsr.2021.105567

[38] Joel Wester, Tim Schrills, Henning Pohl, and Niels van Berkel. 2024. "As an AI Language Model, I Cannot": Investigating LLM Denials of User Requests. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*. 1–14. https://doi.org/10.1145/3613904.3642135

[39] Juliette Zaccour, Reuben Binns, and Luc Rocher. 2025. Access Denied: Meaningful Data Access for Quantitative Algorithm Audits. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. https://doi.org/10.1145/3706598.3713963

[40] J.D. Zamfirescu-Pereira, Jerry Chen, Emily Wen, Allison Koenecke, Nikhil Garg, and Emma Pierson. 2022. Trucks Don't Mean Trump: Diagnosing Human Error in Image Analysis. In *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency (FAccT '22)*. 799–813. https://doi.org/10.1145/3531146.3533145

[41] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric Xing, Hao Zhang, Joseph Gonzalez, and Ion Stoica. 2023. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems 36*. 46595–46623. https://doi.org/10.52202/075280-2020